Fine-Grained Bird Recognition with Annotated Foreground Regions

Zheng Shisheng
The Hong Kong University of Science and Technology
21332987

Lu Yang
The Hong Kong University of Science and Technology
21329126

Abstract

We study how foreground localization, geometric preprocessing, and training strategy affect fine-grained bird recognition on CUB-200-2011. Using an ImageNet-pretrained ResNet-18 and a fixed class-stratified validation split, we compare whole-image inputs, annotated bird boxes processed with center cropping, and bird boxes processed with aspect-ratio-preserving letterboxing. We then isolate the effect of the bird box with a matched whole-image letterbox control, compare frozen and fine-tuned backbones, and ablate the amount of context retained around the bird. The selected model uses the exact annotated bird box, letterboxing, and full fine-tuning from the first epoch. Across three training seeds it reaches $76.06\% \pm 1.54\%$ validation Top-1. The selected seed-43 checkpoint reaches 77.87% Top-1, 94.15% Top-5, and 77.85% Macro-F1 on the official 5,794-image test partition, compared with 77.83% validation Top-1. A background-only diagnostic collapses to 1.61% Top-1, whereas blurring or replacing visible context changes Top-1 by less than 1.2 points. Grad-CAM highlights heads, beaks, body outlines, and wing patterns for successful predictions. The study supports a carefully controlled conclusion: foreground localization is essential in this pipeline, while the utility of background is limited and remains entangled with target scale and retained context. The model requires an annotated bird box.

1. Introduction

Fine-grained recognition asks a model to distinguish visually similar subcategories rather than broad object classes. Birds are a useful test case: species may share an overall silhouette while differing in subtle evidence such as bill shape, head markings, wing coloration, and proportions. CUB-200-2011 provides 200 species, 11,788 images, class labels, bounding boxes, part annotations, and an official train/test division [1]. This setting lets us study recognition quality without collecting a new dataset.

ImageNet-pretrained convolutional networks provide a strong transfer-learning baseline [2,3]. However, a bird classifier can succeed for different reasons. It may use fine-grained foreground evidence, scene context correlated with a species, or accidentally useful image geometry. The practical question in this project is:

> Does focusing the input on the annotated bird region improve fine-grained recognition, and what changes when we alter foreground, context, and training adaptation?

We make four empirical steps. First, we compare whole images, bird center crops, and bird letterbox inputs across three seeds. Second, we hold the letterbox geometry fixed and compare whole-image and bird-region inputs, removing augmentation geometry as an explanation for the main gain. Third, we compare frozen and fine-tuned backbones. Fourth, we ablate the bird-box expansion margin from 0% to 20%, and separately perturb the context visible inside a 10% expanded input.

The final selected model uses the exact CUB box, letterboxing, and full fine-tuning. It reaches 77.87% official test Top-1, consistent with its 77.83% validation result. The project contributes a reproducible benchmark, a controlled demonstration that fine-tuning and foreground localization matter, and an honest account of what background interventions can and cannot establish.

2. Related Work

Transfer learning. ImageNet pretraining remains a strong starting point for smaller vision datasets [3]. Freezing the backbone tests whether existing features are sufficient, while fine-tuning adapts filters to domain-specific evidence. We therefore treat training strategy as an experimental variable rather than assuming that one schedule is uniformly best.

Fine-grained recognition. CUB is a standard benchmark for species, attributes, and part localization [1]. Our work does not propose a new fine-grained architecture. Instead, it isolates the effect of foreground selection and input geometry on a reproducible residual-network baseline.

Visual explanation. Gradient-weighted Class Activation Mapping (Grad-CAM) uses gradients flowing into a convolutional feature map to produce a coarse localization map for a predicted class [4]. We use it to inspect whether successful predictions focus on bird parts and whether failures attend to the wrong regions. Grad-CAM is descriptive evidence, not a causal proof.

3. Data and Problem Statement

CUB-200-2011 contains 200 classes, with 5,994 images in the official training partition and 5,794 images in the official test partition. To keep the test set independent during development, we split only the official training partition. Within each class, images are shuffled with split seed 42 and approximately 20% are assigned to validation. This produces 4,794 training and 1,200 validation images. Every method uses the same image IDs and the identical split hash recorded in each run's environment.json.

The image-level target is one of 200 species labels. We report Top-1 accuracy, Top-5 accuracy, and Macro-F1. Top-1 measures the primary decision, Top-5 measures whether the correct species remains among the leading candidates, and Macro-F1 gives each species equal weight. Three training seeds are summarized with mean and sample standard deviation. These standard deviations characterize training variability and are not confidence intervals.

Every bird-aware condition uses the CUB ground-truth box. This is an additional annotation requirement: the selected method is not an unconstrained end-to-end detector. For a margin m, each side of the box is expanded by m times the original box width or height, clipped to the image boundary. The expanded region is then processed according to the condition.

4. Methods

4.1. Backbone and optimization

We use ResNet-18 with ImageNet-pretrained weights [2]. The final fully connected layer is replaced with a 200-way classifier. Training uses AdamW, weight decay 10^{-4}, batch size 32, mixed precision, and ten epochs. The classifier learning rate is 10^{-3} and the backbone learning rate is 10^{-4}. The checkpoint with the best validation Top-1 accuracy is retained.

Three schedules are compared. Frozen trains only the classifier for all ten epochs, updating 102,600 parameters. Fine-tune updates the backbone from epoch one, totaling 11,279,112 parameters. Warm-up freezes the backbone for the first two epochs and then fine-tunes all parameters. These conditions distinguish feature reuse from domain adaptation.

4.2. Input conditions

Whole image (legacy). Training uses random resized cropping and a horizontal flip. Validation uses resize-to-256 followed by a 224×224 center crop.

Bird center crop. The 10%-expanded bird box is extracted and then processed with the same legacy geometry. This is the original crop baseline.

Bird letterbox. The 10%-expanded bird box is resized so that its longest side is 224 pixels while preserving aspect ratio, then centered on a 224×224 canvas filled with RGB $(124, 116, 104)$. Training adds only a horizontal flip. Validation is deterministic. This avoids discarding a head, tail, or wing when a center crop intersects a narrow region.

Whole-image letterbox control. To avoid confounding bird localization with the removal of random cropping, we apply letterboxing directly to the whole image and compare it with bird-region letterboxing. The two conditions share the network, split, seeds, batch size, and ten-epoch budget.

5. Experiments

5.1. Original input comparison

Table 1 summarizes the original three-way comparison. Whole image reaches $70.31\% \pm 1.13\%$ Top-1, center crop reaches $67.44\% \pm 0.05\%$, and bird letterbox reaches $74.31\% \pm 0.75\%$. Letterbox improves over the whole image by 4.00 points and over center crop by 6.86 points, with the same direction under all three seeds.

The center-crop result is informative: bird localization alone is not sufficient. A narrow crop can remove context and can also clip a head, tail, or other fine-grained evidence. Letterbox preserves the complete expanded region and performs better.

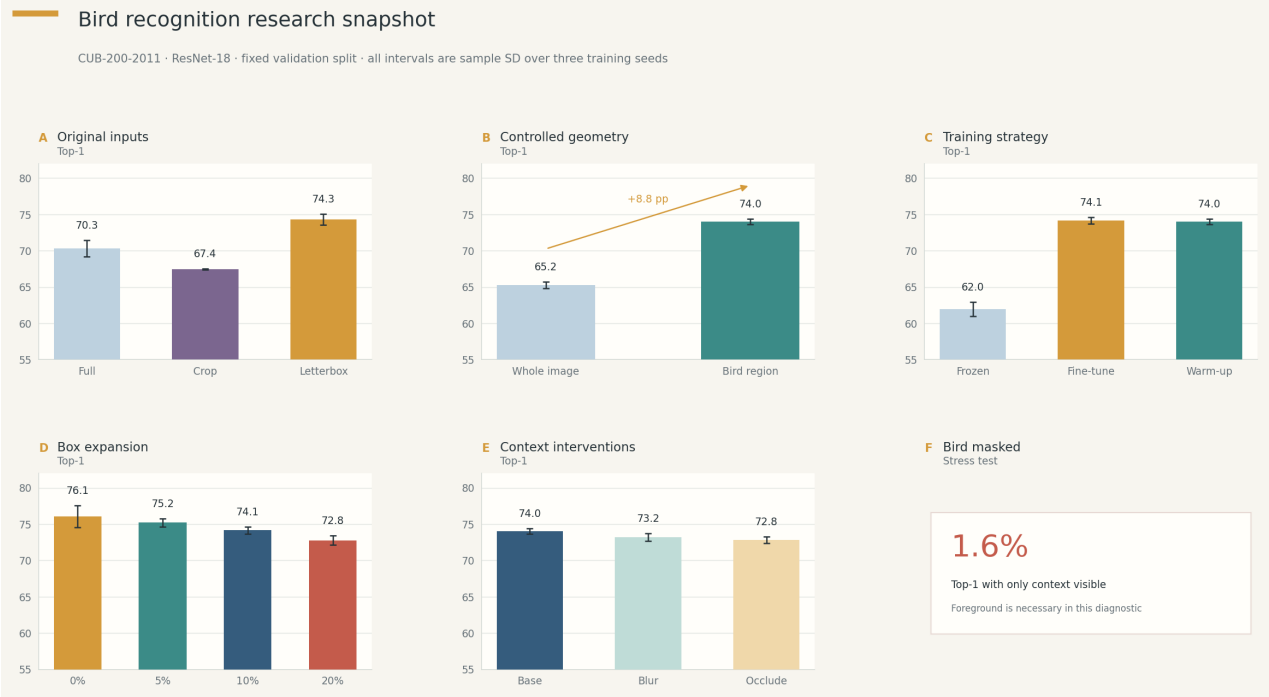


Figure 1. Validation-only research overview. (A) Original input baseline. (B) Matched whole-image versus bird-region letterboxing. (C) Frozen versus fine-tuned training. (D) Box expansion ablation. (E) Context interventions. (F) Bird-masked stress test. Error bars show sample standard deviation over three training seeds.

Validation condition	Top-1	Top-5	Macro-F1
Whole image	70.31±1.13	91.42±0.22	70.02±0.92
Center crop	67.44±0.05	89.64±0.10	67.06±0.23
Bird letterbox	74.31±0.75	93.06±0.42	73.97±0.88
Whole + LB	65.25±0.44	88.28±0.76	64.82±0.55
Bird + LB	74.00±0.36	93.44±0.42	73.91±0.33
Frozen backbone	61.97±0.99	87.86±0.71	61.37±1.20
Fine-tune E1	74.14±0.46	93.08±0.83	73.93±0.59
Freeze 2, tune	74.00±0.36	93.44±0.42	73.91±0.33

Table 1. Validation results (%; mean ± sample SD over seeds 42–44). The lower two blocks are local controls and are not pooled with the original three-way comparison.

5.2. Controlled geometry and training

The matched whole-image and bird-region letterbox experiment produces 65.25% ± 0.44% and 74.00% ± 0.36% Top-1, respectively. The paired improvements are 9.33, 8.08, and 8.83 points for seeds 42, 43, and 44. Because geometry and augmentation are held fixed, removal of random cropping cannot by itself explain the gain. The crop still changes target scale and retained context, so we avoid attributing the entire gain to background.

The training comparison shows a larger and more decisive effect. Frozen features reach 61.97% Top-1, while fine-tuning from epoch one reaches 74.14% and the two-epoch warm-up reaches 74.00%. Fine-tuning gains 11.25–13.83 points over the frozen backbone in every seed. Warm-up and immediate fine-tuning differ by only 0.14 point on average, with inconsistent seed-level signs. We therefore use immediate fine-tuning in the final candidate; warm-up is not claimed to provide a stable benefit.

5.3. Bird-box expansion ablation

Figure 2 varies the box expansion while holding the network, letterboxing, split, seeds, and ten-epoch fine-tuning budget fixed. Top-1 is 76.06% ± 1.54% for margin 0%, 75.19% ± 0.54% for 5%, 74.14% ± 0.46% for 10%, and 72.78% ± 0.63% for 20%.

Margins 0% and 5% both outperform 10% under all three seeds, whereas 20% is worse under all three. The average 0% minus 10% difference is 1.92 points, and the 5% minus 10% difference is 1.06 points. The exact box has the best mean, but its seed variation is larger; 5% is the more stable alternative. Margin 0% and 5% differ by only 0.86 point on average, and no claim of statistical significance is made.

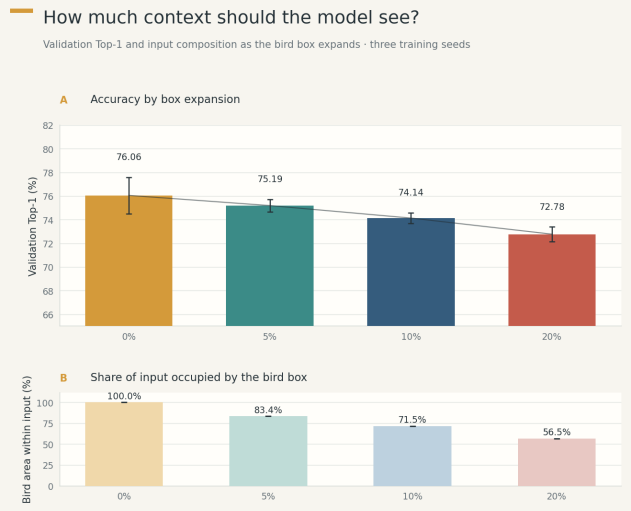


Figure 2. Bird-box expansion ablation. The lower panel reports the mean fraction of the network input occupied by the bird box. Increasing the margin changes target scale, context area, padding, and effective pixels simultaneously.

6. Background Analysis and Interpretation

6.1. Interventions inside the 10% view

We use the three 10%-expanded letterbox models for a validation-only stress test. The baseline result is 74.00% Top-1. Gaussian blur of context outside the tight box changes Top-1 to 73.19%; replacing context with a neutral color changes it to 72.83%. Both directions are consistent across seeds, but the degradation is below 1.2 points. A bird-masked condition that keeps only context collapses to 1.61% Top-1.

Figure 3 shows the same pattern across bird-size groups: context perturbations have limited effect, while masking the bird causes near-complete failure. The smallest group contains only 33 validation images and is exploratory. The evidence supports foreground dependence in normal prediction, but the bird-masked test is a distribution shift and does not prove that background has no role.

6.2. Grad-CAM

Figure 4 shows representative correct predictions and high-confidence failures for the selected seed-43 model. Correct examples typically place attention on the bird's head, bill, torso, or wing patterns. Failures can place attention on broad body regions or surrounding structure without isolating the cue needed to separate similar species. The selected cases are explanatory rather than a random sample, and Grad-CAM is only a coarse visualization.

Partition	Top-1	Top-5	Macro-F1
Validation (1,200)	77.83	93.42	77.63
Official test (5,794)	77.87	94.15	77.85

Table 2. Frozen-model results (%). Model selection used validation only.

7. Final Frozen Model and Test Evaluation

The final model was selected using validation evidence only: margin 0% achieved the best mean, and seed 43 was the best checkpoint within that condition. The model, preprocessing, split, and checkpoint were copied into a frozen directory before the official test partition was opened.

Table 2 reports the single post-freeze evaluation. Top-1 is 77.87%, Top-5 is 94.15%, and Macro-F1 is 77.85% on 5,794 images. The validation Top-1 of the same checkpoint is 77.83%. The 0.04-point test difference is small and descriptive; no significance test is inferred. No checkpoint, seed, margin, or hyperparameter was changed after observing the test results.

Figure 5 compares validation and test metrics. The consistency between partitions suggests that the validation protocol was reasonably representative. Per-class test accuracy has median 83.33% and ranges from 23.33% to 100%, showing that aggregate accuracy hides difficult species.

Figure 6 shows the most frequent official-test confusion pairs. Many errors link visually similar species: shrikes, crows and ravens, terns, flycatchers, nightjars, and hummingbirds. These are plausible fine-grained confusions rather than random label failures.

8. Discussion and Limitations

The main empirical result is that both foreground localization and backbone fine-tuning are necessary in this project. Frozen features perform poorly despite a strong ImageNet initialization. Bird-region inputs outperform whole-image inputs under matched geometry, and tighter boxes outperform larger context windows. The background-only stress test strongly suppresses recognition, while context blur and occlusion have small aggregate effects. Together, these results support a foreground-centered interpretation.

The evidence nevertheless has important limits. First, all bird-aware methods use ground-truth CUB boxes. The selected system is not an automatic detector and requires a box at inference. Second, changing the margin simultaneously changes target scale, context, padding, and effective pixels; it is not a pure background manipulation. Third, context perturba-

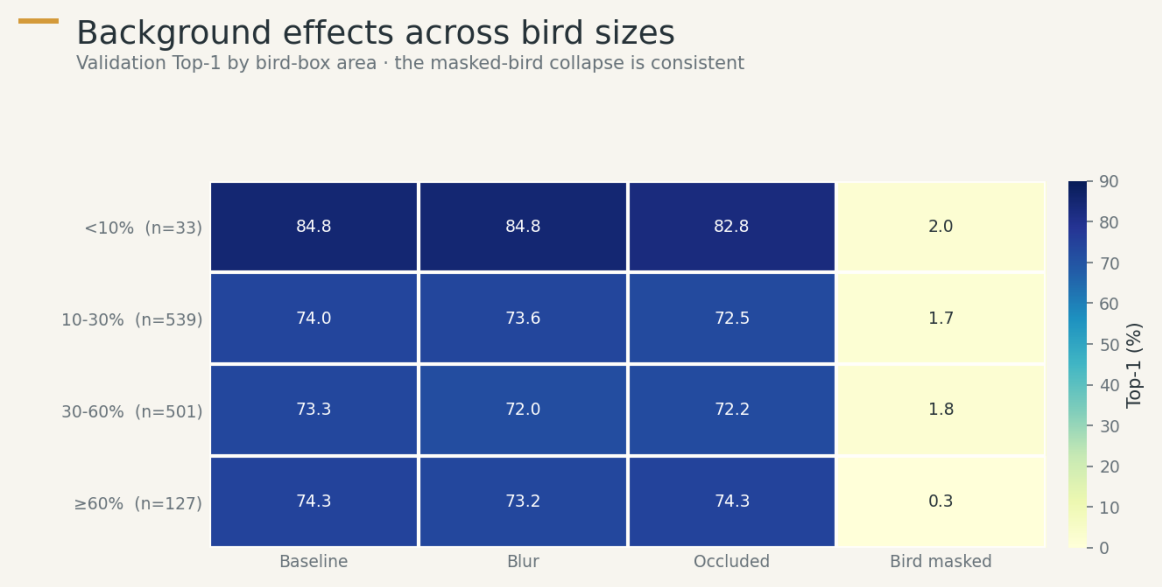


Figure 3. Validation Top-1 by bird-box area under four context conditions. Context blur and occlusion cause small changes; masking the bird causes a large failure in every size group.

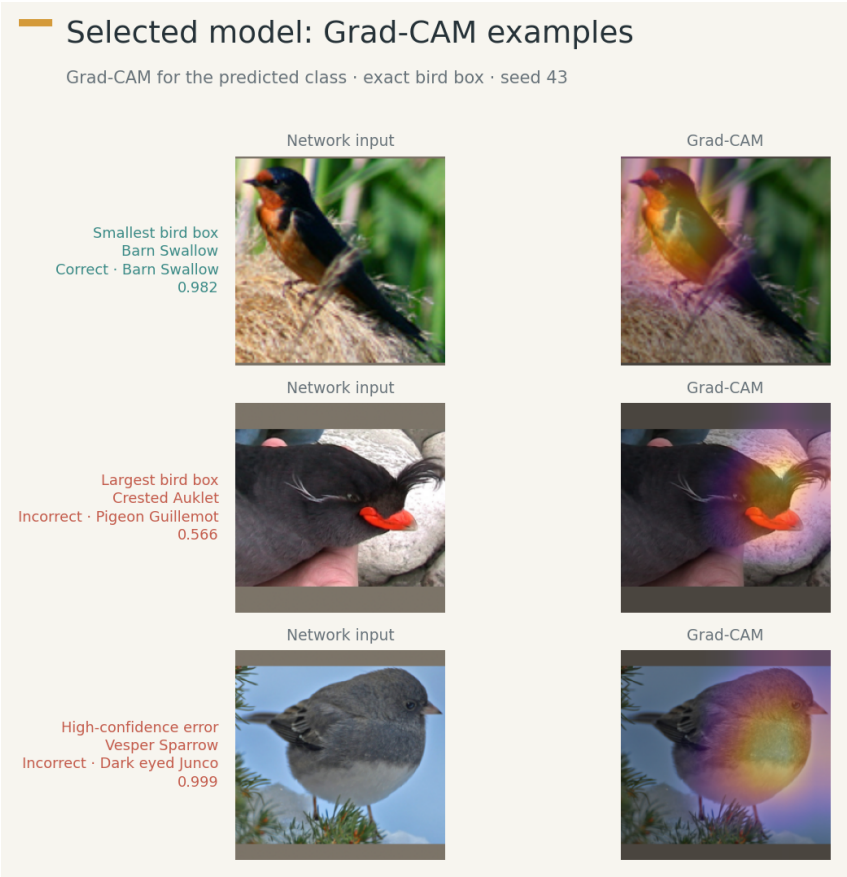


Figure 4. Grad-CAM for the selected margin-0% model. A smallest-box success, largest-box failure, and high-confidence failure illustrate head/body attention and important failure modes.

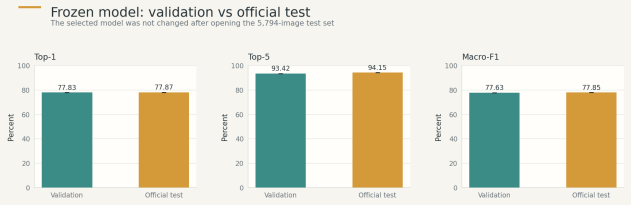


Figure 5. Selected frozen model on validation and the single official test evaluation. The model was not modified after test evaluation.

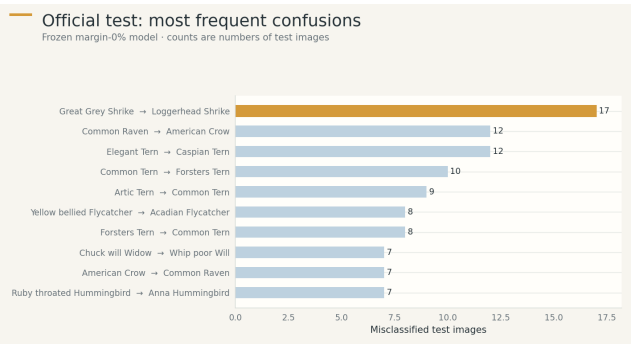


Figure 6. Most frequent official-test confusion pairs for the frozen model.

tions are inference-time distribution shifts rather than retraining experiments. Fourth, CUB may overlap with ImageNet, so pretraining exposure is a shared confound across all conditions. Fifth, only ResNet-18 is tested, and three seeds provide stability evidence rather than a general claim. Finally, probabilities are not calibrated, and Grad-CAM is not a strict causal method.

The 77.87% test Top-1 is a result for a 200-class benchmark with exact localization annotations, not an unconstrained wild-bird identification system. A practical deployment would need either manual boxes or an independent detector, and its errors should be evaluated under that full pipeline.

9. Conclusion

We presented a controlled study of foreground and input geometry for fine-grained bird recognition. Exact bird boxes with aspect-ratio-preserving letterboxing outperform whole-image and center-crop baselines; the gain persists under matched geometry; and full fine-tuning is substantially better than frozen features. The selected model reaches 77.87% Top-1 and 77.85% Macro-F1 on the official test set. Background perturbations inside the bird-centered view cause only small changes, whereas masking the bird causes near-complete failure. Future work should compare automatic detection plus recognition, test additional backbones and augmentation policies, calibrate uncertainty, and evaluate whether the same conclusions hold on bird images outside CUB.

Implementation and AI Disclosure

The implementation, experiment orchestration, and figure-generation code were developed with iterative assistance from Codex. The authors reviewed the data splits, training protocol, evaluation logic, metric calculations, and reported limitations. Public datasets, pretrained weights, libraries, and prior work are cited in the references.

References

[1] C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie. The Caltech-UCSD Birds-200-2011 Dataset. CNS-TR-2011-001, 2011.

[2] K. He, X. Zhang, S. Ren, and J. Sun. Deep Residual Learning for Image Recognition. In CVPR, 2016.

[3] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. ImageNet: A Large-Scale Hierarchical Image Database. In CVPR, 2009.

[4] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. In ICCV, 2017.

[5] PyTorch. Transfer Learning for Computer Vision Tutorial. https://docs.pytorch.org/tutorials/beginner/transfer_learning_tutorial.html